

Second-Hand Product Escrow Marketplace

Problem Statement #39

Exception Flow

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1. Overview

An **exception flow** describes what happens when a use case cannot proceed along its basic flow because of an error, failure, or invalid condition. Each exception flow specifies the **trigger**, **system response**, and **resulting state**.

UC-01: Fund Order into Escrow

E1 – Payment Declined

- **Trigger:** The Payment Gateway declines the transaction due to insufficient funds, an invalid card, or bank refusal.
- **System Response:** The system displays the decline reason, keeps the order in Pending Payment, and does not create an escrow lock. The Buyer may retry using a different payment instrument.
- **Resulting State:** The order remains Pending Payment and does not proceed to Under Inspection.

E2 – Gateway Timeout / Network Failure

- **Trigger:** The connection to the Payment Gateway times out or fails during the transaction.
- **System Response:** The system voids or rolls back any partial authorization to prevent duplicate or partial escrow locks and prompts the Buyer to retry.
- **Resulting State:** The order remains Pending Payment. A reconciliation check confirms that no orphaned charge exists.

UC-02: Submit Inspection Report

E1 – Missing Evidence on Submit

- **Trigger:** The Agent attempts to submit the inspection report without at least one photo or without specifying a pass/fail result.
- **System Response:** The system rejects the submission and displays a validation message. The report remains editable and is not saved as a completed report.
- **Resulting State:** The order remains Under Inspection until a valid inspection report is submitted.

E2 – Inspection Cannot Be Completed

- **Trigger:** The Agent cannot access or reach the item within the scheduled inspection window, such as when the Seller is unavailable.
- **System Response:** The system logs a **Missed Inspection** event and escalates the order to the Admin for rescheduling.
- **Resulting State:** The order remains **Under Inspection / Escalated** and the escrowed funds remain locked.

UC-03: Review and Approve/Reject Item

E1 – No Response Within Window (Fail Result)

- **Trigger:** The Buyer does not respond within 48 hours and the inspection outcome is **Fail**.
- **System Response:** The system does not automatically release the funds. Instead, it flags the order for mandatory Admin manual review because automatically releasing funds after a failed inspection would be unsafe.
- **Resulting State:** The order enters **Pending Admin Review** and the escrow remains locked until the Admin takes action.

E2 – Late Decision Attempt

- **Trigger:** The Buyer attempts to approve or reject the item after the 48-hour decision window has closed, and the order has already been automatically processed because the inspection passed and the Buyer did not respond in time.
- **System Response:** The system rejects the late input, informs the Buyer that the decision window has closed and an outcome has already been applied, and logs the attempt for audit purposes.
- **Resulting State:** The order state remains unchanged because the automatic release has already been applied.

UC-04: Release Escrow Funds

E1 – Payout API Failure

- **Trigger:** The Payment Gateway payout request fails or returns an error during the release process.
- **System Response:** The system retries the payout using a backoff strategy for a limited number of attempts. If all attempts fail, the system marks the order as **Release Failed** and escalates it to the Admin for manual settlement.
- **Resulting State:** The funds remain safely held in escrow. No funds are lost or paid twice.

E2 – Dispute Raised During Release Window

- **Trigger:** A dispute (UC-05) is raised after approval or an automatic release trigger but before the payout is confirmed as settled, creating a race condition.
- **System Response:** The system immediately halts the pending release, in accordance with NFR-001, and routes the order to the dispute-resolution queue instead of completing the payout.
- **Resulting State:** The order moves to Under Dispute. The release either resumes or is cancelled depending on the dispute outcome.

UC-05: Raise Dispute

E1 – Dispute Raised After Funds Are Released

- **Trigger:** The Buyer attempts to raise a dispute after the 48-hour automatic release has already completed and the funds have been transferred to the Seller.
- **System Response:** The system rejects creation of an in-app dispute ticket because the escrow no longer holds the funds. The Buyer is directed to the platform's manual support escalation channel.
- **Resulting State:** No new escrow freeze is created. The case is handled through the manual support process.

E2 – Ticket Creation Failure

- **Trigger:** A backend or system error occurs while creating the dispute ticket.

- **System Response:** As a fail-safe consistent with **NFR-001**, the system keeps or forces the escrow to remain frozen rather than allowing the funds to be released. The error is logged and the Admin is alerted directly.
- **Resulting State:** The funds remain **Locked in Escrow** until the dispute ticket is successfully created or the Admin manually intervenes.